

COMMENTS: EM-BASED LIKELIHOOD INFERENCE FOR SOME LIFETIME DISTRIBUTIONS BASED ON LEFT TRUNCATED AND RIGHT CENSORED DATA AND ASSOCIATED MODEL DISCRIMINATION

Hideki Nagatsuka

Department of Industrial and Systems Engineering, Chuo University
e-mail: *hideki@indsys.chuo-u.ac.jp*

Left truncation and right censoring often occur in the follow-up studies in various fields, such as life-testing, reliability, biology and medical science. Left truncation gives biased sample, while right censoring produces a type of incomplete data. These issues should be resolved, and so developing statistical methodology based on left truncated and right censored data is very important. However, there is relatively little literature dealing with estimation based on left truncated and right censored data, compared with the literature dealing with estimation based on only censored data. Professors Balakrishnan and Mitra are to be congratulated for this excellent and interesting article (Balakrishnan and Mitra, 2014) addressing EM-based inference based on left truncated and right censored data. One of the important contributions of this article is to provide us elegantly derived EM algorithm steps for most commonly used lifetime distributions, involving the lognormal, Weibull and gamma distributions, and their generalization, the generalized gamma distribution, based on the data design by Hong, Meeker and McCalley (2009). It also discusses asymptotic confidence intervals of the parameters based on the asymptotic normality and bootstrap method. In the context of the asymptotic variance-covariance matrices in the asymptotic normality, the observed information matrices and the Fisher information matrices for the four distributions are nicely derived by using the missing information principle. This article further addresses prediction of lifetime in the case of the lognormal distribution, a model discrimination problem based on Akaike's AIC and Schwarz's BIC, and the future work.

I would like to focus on a topic of point estimation of the parameters, in this discussion. In the simulation study for the Weibull distributions, the authors observed that, for $n = 100$, the EM algorithm failed to converge relatively often when $\alpha = 40$ and $\eta = 4$. It is expected that the method fails to converge more often for smaller sample size. It may occur due to instability of the likelihood-based estimation for large η such as 4, and truncation. The larger η becomes, the more unstable the likelihood-based estimation of η becomes, which is roughly confirmed from the fact that the Fisher information of η at the maximum of the likelihood function is decreasing with respect to η . Let us note that Eq.(5.5) of Balakrishnan and Mitra (2014) can be rewritten as

$$Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)}) = Q_{\text{RC}}(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)}) + P_{\tau}(\boldsymbol{\theta}),$$

where $Q_{\text{RC}}(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)})$ is the Q-function when the observed lifetime data of power transformers in-

stalled before 1980 is treated as not left truncated and right censored data but only right censored data, and $P_{\tau}(\boldsymbol{\theta}) = \sum_{i=1}^n (1 - v_i) \{ -(\tau_i^L - \mu)/\sigma + \exp((\tau_i^L - \mu)/\sigma) \}$ (the term $\sum_{i=1}^n (1 - v_i) \exp((\tau_i^L - \mu)/\sigma)$ in Eqs.(5.1) and (5.5) of Balakrishnan and Mitra (2014) might be $\sum_{i=1}^n (1 - v_i) \{ -(\tau_i^L - \mu)/\sigma + \exp((\tau_i^L - \mu)/\sigma) \}$). So, the EM algorithm based on $Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)})$ can be viewed as a penalized EM algorithm based on right censored data, with the penalized term $P_{\tau}(\boldsymbol{\theta})$. We note that $P_{\tau}(\boldsymbol{\theta})$ is strictly decreasing with respect to σ , for any fixed μ , which indicates that the estimate of σ obtained by the method based on $Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)})$ tends to be smaller than the estimate obtained by the method based on $Q_{RC}(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)})$. From these facts, we can say roughly that estimation based on $Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)})$ is more likely to be unstable than that in the case where the observed data of power transformers installed before 1980 is treated as only right censored data.

For improving this instability problem, I consider a substitute penalized function. Now, let us treat the observed and unobserved lifetime data of power transformers installed before 1980 as the right censored data and missing values, respectively. Let w and ζ be missing variates of lifetime and the number of the missing observations, respectively. It is supposed that ζ has density g with respect to a σ -finite measure λ , which is not depending on parameters μ and σ , and the missingness is MCAR. In these setting, the Q-function in an EM framework is given by

$$Q_2(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)}) = Q_{RC}(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)}) + P_2(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)}),$$

where

$$\begin{aligned} P_2(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)}) &= \int \int \zeta \left\{ -\log \sigma + \frac{w - \mu}{\sigma} - \exp\left(\frac{w - \mu}{\sigma}\right) \right\} \\ &\quad \times \frac{1}{\sigma_{(k)}} \exp\left\{ \frac{w - \mu_{(k)}}{\sigma_{(k)}} - \exp\left(\frac{w - \mu_{(k)}}{\sigma_{(k)}}\right) \right\} g(\zeta) dw d\lambda(\zeta). \end{aligned}$$

At least, $P_2(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)})$ is not always decreasing with respect to σ . Let me further assume that we know or estimate the maximum number of the missing observation, ζ_{max} , and select g as discrete uniform pmf on $[0, \zeta_{max}]$ (this distribution may be appropriate when there is no information other than ζ_{max} , as it is the maximum entropy distribution on $[0, \zeta_{max}]$), then $P_2(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)})$ becomes

$$\begin{aligned} P_2(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)}) &= \gamma \int \left\{ -\log \sigma + \frac{w - \mu}{\sigma} - \exp\left(\frac{w - \mu}{\sigma}\right) \right\} \\ &\quad \times \frac{1}{\sigma_{(k)}} \exp\left\{ \frac{w - \mu_{(k)}}{\sigma_{(k)}} - \exp\left(\frac{w - \mu_{(k)}}{\sigma_{(k)}}\right) \right\} dw, \end{aligned}$$

where $\gamma = (\zeta_{max} + 1)/2$. Figure 1 displays the plots of the penalized functions $P_{\tau}(\boldsymbol{\theta})$ (solid) and $P_2(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)})$ (dashed), based on the simulated data from Balakrishnan and Mitra (2012), which is obtained from the extreme value distribution with $\mu = 3.55$, $\sigma = 0.33$, $n = 100$ and truncation percentage 40. $P_2(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)})$ depends on $\boldsymbol{\theta}^{(k)}$, and then, $\sigma^{(k)}$ is taken as 0.33 in the left panel of Figure 1, while $\sigma^{(k)} = 0.7$ in the right panel. $\mu^{(k)}$ is set to 3.55 in both panels. The maximum number of the missing observation ζ_{max} is selected as 120 which is three time as large as the number of the observed data of power transformers installed before 1980. Figure 1 shows that $P_2(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)})$

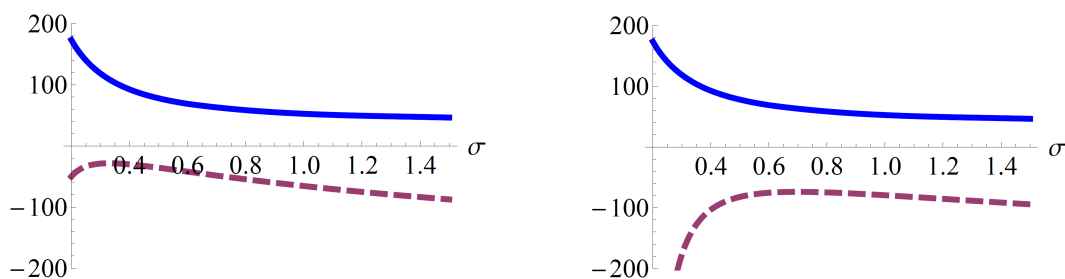


Figure 1: Graph of the penalized functions $P_{\tau}(\boldsymbol{\theta})$ (solid) and $P_2(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)})$ (dashed), based on data from Balakrishnan and Mitra (2012); left panel: $\sigma^{(k)} = 0.33$; right panel: $\sigma^{(k)} = 0.7$.

is unimodal, maximized at around $\sigma^{(k)}$, and increasing when σ is small, while $P_{\tau}(\boldsymbol{\theta})$ is strictly decreasing over $\sigma > 0$ as I mentioned earlier. Let us recall that the smaller σ (the larger η) becomes, the more unstable the likelihood-based estimation becomes, and thus, it is expected that the method of estimation based on $Q_2(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)})$ is more stable than that based on $Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)})$. However, of course, properties of estimators such as bias and MSE should be investigated.

In this method, ζ_{max} should be chosen. It may be estimated based on some historical information, or expert opinions in application. It can also be chosen using cross-validation, AIC, BIC, or other model selection methods even if these are unavailable.

The focus of this discussion was on the EM algorithm. However, this discussion can be applied to the maximum likelihood method. The instability estimation problem for large shape parameter also occurs in the lognormal, gamma and generalized gamma distributions, and so it would be interesting to discuss the problems in them.

References

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